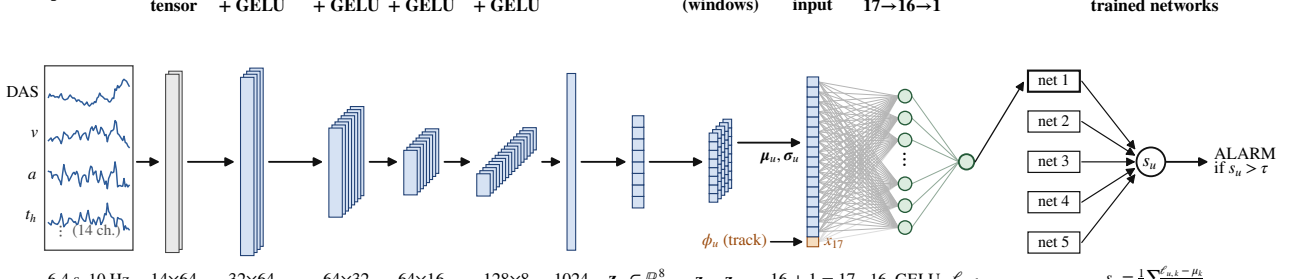
Input window
Input tensor
Conv1d + GELU
Conv1d + GELU
Conv1d + GELU
Conv1d + GELU
Flatten
Linear
Mean ⊕ std (windows)
Head input
MLP 17→16→1
Ensemble of 5 trained networks
DAS
v
a
t_h
(14 ch.)
net 1
net 2
net 3
net 4
net 5
s_u
ALARM if $s_u > \tau$
μ_u, σ_u
ϕ_u (track)
x_{17}
6.4 s, 10 Hz
14×64
32×64
k=7, s=1
64×32
k=5, s=2
64×16
k=5, s=2
128×8
k=3, s=2
1024
$\mathbf{z}_w \in \mathbb{R}^8$
1024→8
$\mathbf{z}_1 .. \mathbf{z}_W$
16 + 1 = 17
16, GELU
$\ell_{u,k}$
$s_u = \frac{1}{5}\sum_k \frac{\ell_{u,k} - \mu_k}{\sigma_k}$
$\tau = 1.973$ (frozen)
active (as trained; all main results): $x_{17} = (\phi_u - \mu_\phi)/\sigma_\phi$
neutralization check (post hoc): $x_{17} = 0$